

SNLP Course work 1

Introduction

The Primary objective of this course work is Natural Language Processing (NLP) to do an analysis on the given input text biography. When I run text through the spaCy's library, it turns the input text into a form of structured data it can also be use to find more about words and grammar. The first step in the process is tokenisation, which shows that there are 302 words and 176 distinct phrases. The average length of words are 5.66 characters and the average sentence length is 18.88 words. Named Entity Recognition (NER) is an advanced technology that find's organisations like Edinburgh Napier University, SR University and Syntactic Parsing can map grammatical connections. Lastly, the extraction of 755 n-gram characteristics shows how complicated the language is in the professional introduction. This work reveals the effectiveness of NLP techniques in the process of transformation of personal history into quantifiable technical profiles.

Methodology

The analysis passed through a sequence of computational linguistics:

- Data Loading/Preprocessing: This was done by applying the `en_core_web_sm` pipeline of spacy to the text, which had been read in the form of a file named `snlp_cw1_pragnya.txt`.
- Lexical Analysis: The input was tokenised to lower case words, with no spaces or punctuation.
- Structural Analysis: The based on the SpaCy sentence boundary detection the material was separated in distinct sentences.
- POS Tagging: Labels a set of pre-determined grammatical categories (nouns, verbs, and so on) to each token of the semantic and syntactic analysis.
- NER: Extraction of places, organisations, and names of people.
- Dependency parsing: the grammatical relations among words (subject, object or root) were displayed and traversed.
- Features Extraction: Word sequence frequencies, N-grams (unigrams, bigrams and trigrams) were generated using the NLTK library.

Result

Key Text Statistics

Metric	Result
Total Word Count	302 words
Unique Word Count	176 words
Average Word Length	5.66 characters
Total no. of Sentences	16
Average Sentence Length	18.88 words

Linguistics Observations:

- The most frequent Words: And, I, my and data are the most frequent words.
- Pos Distribution: The most common labels were "PROPN" (Proper Nouns) and "NOUN" (Nouns) which showed the extent of description of the biographical writing. The model had treated and correctly defined outputs.
- NER Accuracy: The model was able to identify Edinburgh Napier University as an ORG and Scotland as a GPE. However, it identified different technical words including C++ and Java in the PERSON group.
- N-gram Features: There were 755 total unique characteristics (unigrams, bigrams and trigrams)

Conclusion

I conclude, the course work was a good demonstration of the practical application of natural language processing concepts when analysing a text. Dependency parsing is a rule-based study of the grammatical structure of sentences, which explains the syntactic structure and hierarchical word dependence. The unigram, bigram and trigram extraction gave the text a statistical representation of feature dimensionality of higher order n-grams. This is because when these methods are combined, they reveal both structural and distributional perspectives when analysing the language. The Python implementation proves the effectiveness the process used linguistic analysis and improves the understanding of the modern text processing techniques.